

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to VTU, Belagavi - 590018)

Accredited by National Assessment & Accreditation Council (NAAC) with 'A' grade

Shavige Malleshwara Hills, Kumaraswamy Layout

Bengaluru-560111



Mini Project Report on

Flag hoisting using IoT

Submitted in partial fulfillment for the award of degree of

**Bachelor of Engineering
in
Electrical and Electronics Engineering**

Submitted by

RAJA BABU RAY

SIDDESHWAR M S

VARDHAN M

PRADYUMNA G NADIGER

1DS23EE074

1DS23EE100

1DS23EE117

1DS24EE408

Under the Guidance of

Dr. Soni M

Assistant professor

Dept. of E&E Engg.

DSCE, Bengaluru

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANASANGAMA, BELAGAVI-590018

2024-25

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to VTU, Belagavi – 590018, Approved by AICTE & ISO 9001:2015 Certified)
 Accredited by National Assessment & Accreditation Council (NAAC) with 'A' grade
 Shavige Malleshwara Hills, Kumaraswamy Layout
 Bengaluru-560111
 2024-2025

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING



CERTIFICATE

Certified that the mini project report entitled “Flag hoisting using IoT” carried out by **RAJA BABU ROY, SIDDESHWAR M S, VARDHAN M, PRADYUMNA GN** bearing **USN: 1DS23EE074, 1DS23EE100, 1DS23EE117, 1DS24EE408** are bonafide students of **DAYANANDA SAGAR COLLEGE OF ENGINEERING**, an autonomous institution affiliated to VTU, Belagavi in partial fulfillment for the award of Degree of **Bachelor of Engineering in Electrical and Electronics Engineering** during the year **2024-2025**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The mini project report has been approved as it satisfies the academic requirements in respect of work prescribed for the said Degree.

Signature of the Guide

Dr. Soni M
 Assistant Professor
 Dept. of E&E Engg.
 DSCE, Bengaluru

Signature of the HOD

Dr. Premkumar M.
 Professor & HOD
 Dept. of E&E Engg.
 DSCE, Bengaluru

Name of the Examiners

1.

2.

Signature with date

.....

.....

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to VTU, Belagavi – 590018, Approved by AICTE & ISO 9001:2015 Certified)
Accredited by NBA, National Assessment & Accreditation Council (NAAC) with 'A' grade
Shavige Malleshwara Hills, Kumaraswamy Layout
Bengaluru-560111
2024-2025

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING



DECLARATION

We , Raja babu Ray (1DS23EE074), Siddeshwar M S (1DS23EE100), Vardhan M (1DS23EE117) and Pradyumna G Nadiger (1DS24EE408), respectively, hereby declare that the mini project work entitled “ Flag hoisting using IoT ” has been independently done by us under the guidance of ‘Dr. Soni M, Assistant Professor , EEE department and submitted in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Electrical & Electronics Engineering, at Dayananda Sagar College of Engineering, an autonomous institution affiliated to VTU, Belagavi during the academic year 2024-2025.

We further declare that we have not submitted this report either in part or in full to any other university for the award of any degree.

**RAJA BABU ROY
SIDDESHWAR M S
VARDHAN M
PRADYUMNA**

**1DS23EE074
1DS23EE100
1DS23EE117
1DS23EE408**

PLACE: Bengaluru

DATE:26/05/2025

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of people who made it possible and under whose constant guidance and encouragement the task was completed. We express our sincere regards to the **Management of Dayananda Sagar College of Engineering, Bengaluru.**

We express our sincere regards and thanks to **Dr. B G Prasad, Principal, Dayananda Sagar College of Engineering, Bengaluru.** His incessant encouragement, guidance and valuable support have been an immense help in realizing this mini project.

We express our sincere regards and thanks to **Dr. Premkumar M, Professor & HOD, Department of Electrical and Electronics Engineering, Dayananda Sagar College of Engineering, Bengaluru.** His incessant encouragement, guidance and valuable technical support have been of immense help in realizing this mini project. His guidance gave us the environment to enhance our knowledge, skills and to reach the pinnacle with sheer determination, dedicated and hard work.

We would like to express profound gratitude to our guide **Dr. Soni M, Assistant professor, Department of Electrical and Electronics Engineering, Dayananda Sagar College of Engineering, Bengaluru** who has encouraged us throughout the mini project. Her moral support enabled us to complete our work successfully.

We express our sincere thanks to Mini Project Coordinator **Dr. Suchana Mishra, Assistant Professor, Department of Electrical and Electronics Engineering** for her continued support and guidance. We thank all teaching and non-teaching staff of the Electrical and Electronics Engineering department for their kind and constant support throughout the academic Journey.

RAJA BABU RAY

1DS23EE074

SIDDESHWAR M S

1DS23EE100

VARDHAN M

1DS23EE117

PRADYUMNA G NADIGER

1DS24EE408

TABLE OF CONTENT

Description	Page No.
Chapter 1:	
CERTIFICATE	i
DECLARATION	ii
ACKNOWLEDGEMENT	iii
TABLE OF CONTENTS	iv - v
LIST OF FIGURES	vi
LIST OF TABLES	vii
CO-PO MAPPING SHEET	viii
ABSTRACT	ix
Chapter 1:	
1.1 Introduction	
1.2 Background	
1.3 Relevance	
1.3.1 Problem Statement	
1.3.2 Objectives	
1.3.3 Scope of the project	
1.3.4 Motivation	
Chapter 2:	
2.1 Literature Survey	
Chapter 3:	System Analysis
3.1 System Architecture	
3.2 Functional Specifications	
3.3 Components	
3.4 Block diagram	
3.5 Circuit diagram	

Chapter 4: Implementation

4.1 Simulation setup
4.3 Software/Embedded system development
4.4 Description of working
4.5 Challenges

Chapter 5: Result and Discussion

5.1 Testing of Prototype
5.2 Observations
5.3 Prototype

Chapter 6: Applications and Advantages

6.1 Real world applications
6.2 Benefits of the system
6.3 Industrial relevance
6.4 Sustainable development goal

Chapter 7: References

7.1 Conclusion
7.2 References

LIST OF FIGURES

Chapter 3:

Figure 3.2 Block diagram

Figure 3.4 Circuit diagram

Chapter 5:

Figure 5.3 Prototype

LIST OF TABLES

Chapter 3:

Table 3.1 Comparison of sensors



DAYANANDA SAGAR COLLEGE OF ENGINEERING
 (An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
 Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001-2015 Certified
 Institution)

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

Department of Electrical and Electronics Engineering

(Accredited by NBA Tier 1: 2022-2025)

CO-PO Mapping

COURSE OUTCOMES

CO1: Identify a research problem by doing literature survey with supervisor's advice.

CO2: Apply the fundamental knowledge of mathematics, science and engineering principles in design of solutions of system components.

CO3: Select and apply a suitable engineering/IT tool in modeling/data interpretation /analytical studies, conduct experiments leading to a logical solution.

CO4: Design a system/ system component/process, build it and test its functioning as a solution to a complex engineering problem with sustainable practice.

CO5: Communicate effectively to a diverse audience about the outcome and societal impact; develop technical reports and publication.

CO-PO mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PS12	PS O1	PS O2	PS O3
CO1		2											2	2	1
CO2	1												2	2	1
CO3	2				2								2	2	1
CO4		3	3	3	3	3	3						2	2	1
CO5						3		3		3		3	2	2	1

	PO1	PO2	PO3	PO4	PO5	PO6	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1		2									2	2	1
CO2	1										2	2	1
CO3	2				2						2	2	1
CO4		3	3	3	3	3					2	2	1
CO5						3	3		3	3	2	2	1

ABSTRACT

With the growing adoption of automation and smart technologies, the integration of the Internet of Things (IoT) into ceremonial processes presents new opportunities for innovation. This project focuses on the design and implementation of an IoT-based automated flag hoisting system, aimed at enhancing convenience, safety, and precision. The system utilizes a microcontroller (such as Arduino or ESP32) connected to a motorized pulley mechanism, enabling remote control via a mobile app or web interface through Wi-Fi or Bluetooth. Sensors such as rotary encoders or limit switches ensure accurate positioning and prevent over-rotation or mechanical failure. Additionally, features such as scheduled hoisting, weather condition alerts, and status notifications are incorporated to optimize operation. This system is particularly useful in institutions, schools, military bases, or remote areas where manual hoisting may be impractical or unsafe. The proposed solution not only automates a traditional task but also demonstrates the applicability of IoT in enhancing everyday civic and ceremonial practices.

Keywords: Esp 32 , Rain sensor, Universal Motor.

Chapter – 1

1.1. Introduction

Flag hoisting is a ceremonial act of great national importance, symbolizing unity, pride, and patriotism. Traditionally, this task is performed manually, requiring the physical presence of individuals to raise and lower the national flag, especially during events such as Independence Day, Republic Day, or institutional functions. However, in certain scenarios—such as remote locations, adverse weather conditions, or the need for synchronized multi-location events—manual flag hoisting can be challenging or impractical. With the rapid advancement of technology, the Internet of Things (IoT) has emerged as a powerful tool for automating everyday tasks by connecting devices to the internet and enabling remote control and monitoring. This project explores the application of IoT in automating the flag hoisting process. By using a microcontroller (such as an ESP32 or Arduino with Wi-Fi capability), a motor-driven pulley system, and various sensors (e.g., limit switches, encoders), the system enables remote, scheduled, and secure operation of flag hoisting via a smartphone or web interface. The proposed system not only ensures precise and safe flag movement but also enhances convenience and accessibility, especially for institutions that wish to automate ceremonial tasks. It supports features such as scheduled flag hoisting based on real-time clocks, sensor-based feedback for system status, and remote monitoring. This project demonstrates how IoT can be leveraged to modernize traditional practices, making them more efficient and adaptive to modern needs.

1.2. Background

Flag hoisting is a deeply symbolic act, often performed during national ceremonies, institutional events, and public gatherings. Traditionally, it requires manual effort, where individuals physically operate a rope-and-pulley system to raise or lower the flag. While this method is effective for small-scale or supervised events, it poses challenges in terms of precision, synchronization, safety, and accessibility—particularly in remote locations, tall flagpoles, or adverse environmental conditions. In recent years, the integration of automation and smart technology into everyday systems has led to increased interest in developing solutions that reduce human effort and increase reliability. The Internet of Things (IoT), a network of interconnected devices capable of collecting and exchanging data, has proven effective in numerous domains, including smart homes, agriculture, healthcare, and industry. By applying IoT to ceremonial processes like flag hoisting, users can remotely manage the system through internet-connected interfaces, ensuring both convenience and efficiency. Previous implementations of automated flag hoisting have largely relied on basic mechanical or manually controlled motor systems without real-time monitoring or remote control. By incorporating IoT capabilities, the system can be enhanced to include features such as scheduled hoisting via real-time clocks, motor control through mobile or web apps, safety mechanisms using sensors, and status notifications. This makes it particularly useful in locations with limited personnel or the need for synchronized hoisting across multiple sites. The development of an IoT-based flag hoisting system represents a step forward in modernizing traditional practices while preserving their cultural significance.

1.3. Relevance

1.3.1. Problem statement

Traditional flag hoisting methods require manual intervention, which can be inconvenient, unsafe, or impractical in certain situations—such as remote areas, tall flagpoles, or during unfavorable weather conditions. Manual systems also lack the precision, scheduling capability, and remote accessibility needed for automated or synchronized events across multiple locations. This limitation becomes especially evident during national ceremonies or institutional events where reliability and timing are critical. There is a need for an automated, remote-controlled, and sensor-monitored flag hoisting system that ensures safety, accuracy, and ease of operation. By leveraging IoT technology, such a system can provide smart control, scheduled operation, real-time feedback, and greater accessibility, thus modernizing and improving the efficiency of the flag hoisting process.

1.3.2. Objectives

The primary objective of this project is to design and implement an IoT-based automated flag hoisting system that enables remote, safe, and precise control of the flag hoisting process. The system aims to modernize traditional flag ceremonies by integrating smart technology for increased convenience and operational efficiency.

Specific objectives include:

1. To develop a motorized pulley system for raising and lowering the flag using a microcontroller (e.g., ESP32, Node MCU).
2. To enable remote control and scheduling of the hoisting process through a mobile or web application using Wi-Fi or internet connectivity.
3. To integrate safety mechanisms using sensors (e.g., limit switches or encoders) to detect flag position and prevent mechanical failure.
4. To implement a real-time clock or time synchronization for automated flag hoisting at scheduled intervals (e.g., during national events).
5. To provide real-time system feedback or notifications to users, confirming task completion or reporting faults.
6. To promote the use of IoT in civic and ceremonial functions, improving accessibility, especially in remote or high-risk environments.

1.3.3. Scope of the Project

1. **Remote Operation:** Enabling users to raise or lower the flag from a smartphone or web application using Wi-Fi or internet connectivity.
2. **Automation & Scheduling:** Incorporating real-time clocks (RTC) or NTP synchronization to automate flag hoisting at predefined times, useful for institutions with recurring flag ceremonies.
3. **Sensor Integration:** Using limit switches, encoders, or IR sensors to monitor the position of the flag and ensure accurate and safe operation, including auto-stop mechanisms.
4. **Motor Control System:** Developing a motorized pulley mechanism controlled by a microcontroller (e.g., ESP32, Arduino) to physically raise and lower the flag.
5. **User Interface:** Creating a user-friendly mobile or web interface that allows manual control, scheduling, and system monitoring.
6. **Notification System (Optional):** Providing real-time alerts or confirmations through SMS, email, or in-app notifications on task completion or system errors.

1.3.4. Motivation

Flag hoisting is a ceremonial tradition that holds deep national and institutional significance. However, the manual nature of traditional flag hoisting poses challenges, particularly in terms of accessibility, safety, and synchronization during large-scale or remote events. In institutions such as schools, government buildings, and military bases, the need for timely and error-free execution is critical-especially during national celebrations. The increasing availability and affordability of IoT devices provide a unique opportunity to automate this important task. The motivation behind this project is to apply IoT technology to enhance the flag hoisting process by making it more efficient, remotely accessible, and less dependent on human effort. The system not only aims to reduce the physical risks and constraints involved but also offers an innovative way to combine tradition with technology, making ceremonial practices smarter and more adaptable to modern needs.

Chapter – 2

2.1.Literature Survey

1. G. Kaur, A. Saini (2020) - “IoT Based Smart Home Automation System”

This paper outlines how IoT can be used to remotely control home appliances using Wi-Fi-enabled microcontrollers. The use of Node MCU, sensors, and relay modules for real-time control demonstrates the feasibility of applying similar architecture for flag hoisting.

(International Journal of Scientific & Engineering Research, Vol. 11, Issue 6)

2. P. S. Kulkarni, A. B. Deshmukh (2019) - “IoT Based Automatic Gate Control System”

This work discusses the use of servo motors and limit switches to control gate operations through an IoT interface. The safety and accuracy techniques used are directly applicable to motorized flag hoisting systems.

(International Research Journal of Engineering and Technology, Vol. 6, Issue 4)

3. M. A. Khan, S. S. Rizvi (2018) - “Design and Implementation of an IoT Based Smart Irrigation System”

Their system uses sensors and microcontrollers to automate and schedule irrigation. The integration of RTC and automation logic mirrors the approach required for timed flag hoisting.

(IEEE International Conference on Advances in Computing and Communication Engineering)

4. S. D. Lokhande, R. N. Awale (2021) - “Smart Bell and Lighting System for Schools Using IoT”

This project implements a real-time clock-based IoT system to automate school bell ringing and lighting. The study emphasizes the use of NTP synchronization and mobile app interfaces, which can be adapted for flag hoisting timing automation.

(IJRASET, Vol. 9, Issue VI)

5. A. Sharma, V. Kumar (2022) - “Remote Monitoring and Control of Industrial Motors Using IoT”

This research focuses on IoT-based motor control systems with real-time feedback and alert mechanisms. The use of MQTT protocols, sensor integration, and remote diagnostics directly informs the technical implementation of flag hoisting mechanisms.

(IEEE Access, Vol. 10)

Chapter-3

System Analysis

3.1. System Architecture

1. User Interface Layer:

Mobile Application or Web Dashboard: Users interact with the system remotely through an app or web interface to issue commands (raise/lower flag), set schedules, and receive notifications about the flag's status.

2. Communication Layer:

Wi-Fi Network & Cloud: The system connects to the internet via Wi-Fi (ESP32/NodeMCU) using MQTT or HTTP protocols to send and receive data between the user interface and the hardware.

3. Controller Layer:

Microcontroller (ESP32/Node MCU): Acts as the central control unit, processing commands from the user interface, controlling the motor driver, reading sensor inputs, and managing scheduled operations via RTC.

4. Hardware Layer:

Motor Driver (L298N): Receives control signals from the microcontroller and powers the DC motor accordingly to raise or lower the flag.

DC Motor & Pulley Mechanism: Converts electrical signals into mechanical motion to physically move the flag up and down.

Sensors (Limit Switches or Rotary Encoders): Detect the position of the flag (top and bottom) to prevent overrun and protect mechanical components.

Real-Time Clock (RTC) Module: Keeps track of time to enable automated, scheduled flag hoisting events.

Power Supply: Provides stable power to all electronic components and motor.

3.2. Functional Specifications

1. Remote Flag Hoisting: The system allows users to raise or lower the flag remotely via a mobile app or web interface using Wi-Fi connectivity.

2. Scheduled Operation: Users can schedule automatic flag hoisting and lowering at specific times using an RTC module or NTP-based clock sync.

3. Motor Control: A DC or servo motor, controlled via a motor driver (e.g., L298N), operates the pulley system for flag movement.

4. Position Detection: Limit switches or rotary encoders detect the flag's upper and lower positions to prevent overrun or damage to the motor.

5. User Interface: A mobile or web dashboard displays options for manual control, schedule setup, and real-time system status.

6. Safety Mechanisms: The system automatically stops the motor when the flag reaches its maximum or minimum position.

In case of motor failure or obstacle detection, the system halts and notifies the user.

7. Real-Time Notifications: The system provides real-time feedback via the user interface (or optional SMS/email) upon successful flag hoisting or fault detection.

8. Internet Connectivity: Uses Wi-Fi (via ESP32 or Node MCU) to communicate with the cloud or user interface for data transmission and control commands.

9. Manual Override (Optional): A physical button can be added as a fallback for manual control in case of internet failure.

10. Data Logging (Optional): Stores logs of operations (e.g., hoisting time, user actions, errors) in a cloud database such as Firebase or Thing speak.

3.3. Components

1. Microcontroller Unit:

ESP32 / Node MCU (ESP8266): Wi-Fi-enabled microcontroller that controls the system, processes inputs, and communicates with the cloud or user interface.

2. Motor & Motor Driver:

DC Motor / Servo Motor / Stepper Motor: Used to drive the pulley mechanism for raising and lowering the flag

Motor Driver Module (e.g., L298N, L293D): Interfaces between the microcontroller and motor to provide adequate voltage and current control.

3. Sensors:

Limit Switches: Detect end positions (top/bottom) of the flag to stop the motor automatically.

Rotary Encoder (optional): Provides precise tracking of flag position or pulley rotation.

4. Real-Time Clock:

RTC Module (e.g., DS3231): Maintains accurate timing for scheduled flag hoisting without needing constant internet connection.

5. Connectivity Modules:

Built-in Wi-Fi (ESP32/Node MCU): Enables communication with a mobile app or web interface over the internet.

6. Power Supply:

DC Power Adapter / Battery Pack / Solar Panel (optional): Provides stable power to the microcontroller and motor system.

7. User Interface (Software):

Mobile App (e.g., Blynk, custom app): Allows remote control, scheduling, and monitoring.

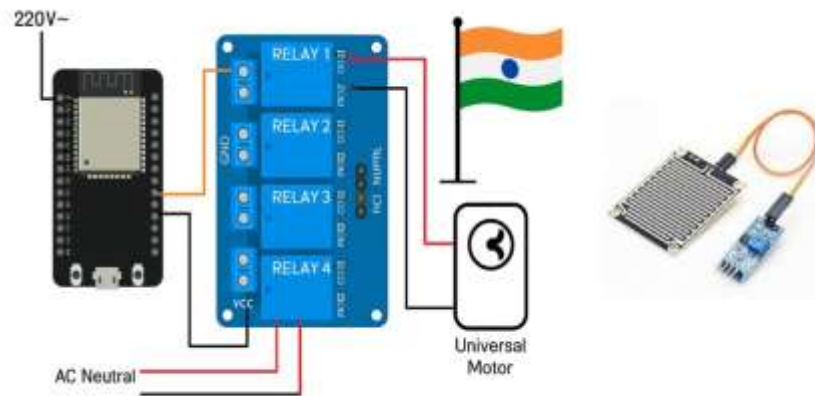
Web Dashboard (e.g., Firebase, HTML/JavaScript): Provides access via browser for control and status updates.

8. Optional Modules:

Buzzer / LED Indicator: Provides audible or visual feedback during operation or errors.

Camera Module (optional): For live visual monitoring of the flag position.

3.4. Block Diagram



3.5. Circuit Diagram



Fig 3.2. Circuit diagram

Chapter-4

IMPLEMENTATION

4.1 Hardware Setup:

ESP32/Node MCU: Acts as the main controller. Connect it to:

RTC Module (e.g., DS3231) via I2C (SDA, SCL)

Motor Driver (L298N) using GPIO pins (IN1, IN2, ENA)

Limit Switches on digital input pins for position detection

Power Supply: 12V for motor, 3.3V/5V for ESP32 and RTC

Motor and Pulley: Connect a DC motor to the L298N OUT1 and OUT2 terminals. Attach a pulley to the motor shaft and connect the flag rope.

Limit Switches: Place one at the top and one at the bottom of the flagpole to detect when the flag reaches the upper or lower limit.

4.2 Software Development:

Programming Platform: Use Arduino IDE to write code for ESP32. Key functions include: Reading time from the RTC, Accepting commands via Wi-Fi, Controlling motor based on command or schedule, Checking limit switch status to stop the motor, Communication Protocol: Use MQTT (e.g., via Adafruit IO) or HTTP (Firebase/Thing Speak) to connect the ESP32 to the cloud interface.

Mobile/Web App: Build a simple app using: Blynk, MIT App Inventor, or custom HTML/CSS/JS for control

Features: Manual flag control, schedule setup, system status, notifications

4.3 Software/Embedded System Development:

Microcontroller: ESP32 or Node MCU used for motor control and IoT connectivity.

Firmware: Developed in Arduino IDE using C/C++.

Functions: Connect to Wi-Fi, Read time from RTC (DS3231), Control motor via L298N, Read limit switches for position detection, Communicate with mobile/web app via MQTT or HTTP, Modes: Manual (via app) and Scheduled (via RTC)

Safety: Stops motor using limit switches and timeout checks.

Output: Real-time updates to user interface and safe automated flag hoisting.

4.4. Description of working

1. User Control or Schedule Trigger: The system can be activated manually via a mobile app or automatically based on a preset schedule using an RTC (Real-Time Clock).
2. Wi-Fi Communication: The ESP32 microcontroller connects to the internet via Wi-Fi and receives commands from the user or checks the current time for scheduled operation.
3. Motor Activation: Based on the input, ESP32 sends control signals to the motor driver (L298N), which powers the DC motor to raise or lower the flag through a pulley mechanism.
4. Position Detection: Limit switches are placed at the top and bottom of the flagpole. When the flag reaches either end, the corresponding switch is triggered, and the motor stops automatically.
5. Status Feedback: The system sends real-time status updates (flag raised, lowered, or errors) back to the user through the app or web dashboard.
6. Safety Features: If the flag does not reach a limit switch within a defined time, the system stops the motor and sends an error alert to prevent damage.

4.5. Challenges

1. Connectivity Issues: Unstable or weak Wi-Fi signals can interrupt remote control and scheduling. IoT commands may fail if there's a delay or loss in internet communication.
2. Power Reliability: Outdoor installations may suffer from power outages or fluctuations. Need for backup power or solar integration in remote areas.
3. Environmental Exposure: Motors, sensors, and microcontrollers are vulnerable to rain, heat, dust, and wind. Requires weatherproof enclosures and robust mechanical design.
4. Motor Control Precision: Incorrect motor control can cause overwinding or slack in the rope. Requires precise calibration and reliable end-stop detection using limit switches.
5. Sensor Failure: Limit switches may malfunction due to dust, rust, or wear. Failure to detect flag position can lead to system damage or safety risks.
6. Time Sync Issues: RTC modules may drift over time if not regularly synced. Leads to incorrect scheduling of flag hoisting events.
7. Security Risks: Without proper encryption, IoT commands can be intercepted or misused. Needs secure authentication for user access.
8. Software Bugs: Faulty logic in firmware can cause flags to stop mid-way or run motors endlessly. Requires thorough testing of all edge cases.

Chapter-5

Results and discussion

5.1. Testing of Prototype

To test the model first, we have listed all the obstacles. During the testing phase of the prototype model schematic evaluation conducted to access its ability to avoid the obstacles. By placing the different objects in front, back left and right sides of the prototype we tested the range and the types of obstacle which is detected by the prototype. We also placed the obstacles all around the obstacle to test detection for 360-degree.

5.2. Observations

By testing came to know that the prototype was unable to detect the objects which are far away from the prototype model. Since the range of the IR sensor is smaller so it was unable to detect the things from far. During the testing phase we observed that it can't be detect the objects which are left or right from the prototype it is only detecting the obstacles which are present right in front of it. These are the prototype physical errors which we observed by the testing. Overall, the rigorous testing process coupled with the diligent debugging errors has enhanced the reliability and function of the prototype module reinforcing its effectiveness in detecting various obstacle in the way to the car.

5.3. Prototype



Fig 5.1. Prototype

Chapter-6

Applications And Advantages

6.1. Real world Applications

1. Educational Institutions:

Automate flag hoisting during national holidays or daily assemblies.

Controlled remotely by school management through a mobile app.

2. Government Buildings:

Ensures timely and consistent flag hoisting in municipal offices, ministries, etc.

Reduces human error and adds security via remote access.

3. Military Bases:

Secure and automated flag operations during specific timings like Reveille and Retreat.

Can be integrated with central command systems for remote monitoring.

4. Smart Cities:

Part of smart public infrastructure, automated during important events or festivals.

Integrated with city IoT systems for scheduling and maintenance alerts.

5. Heritage and Tourist Sites:

Flags hoisted automatically at historical landmarks to enhance ceremonial value.

Reduces the need for manual staff presence at odd hours.

6.2. Benefits of the system

1. Automation:

Automatically hoists and lowers the flag based on scheduled time using an RTC.

Reduces the need for manual intervention.

2. Remote Operation:

Enables control from anywhere using a smartphone or web interface.

Useful for managing multiple locations.

3. Precision and Safety:

Stops automatically at top and bottom positions using limit switches.

Avoids damage due to overwinding or human error.

4. Time Accuracy:

Ensures flag hoisting happens exactly at the scheduled time, especially on national days.

5. Real-Time Monitoring:

Sends live status updates and alerts to users.

Helps in maintenance tracking and quick fault detection.

6. Energy Efficient:

System runs only when needed, conserving energy.

Can be powered by solar systems for outdoor independence.

7. Accessibility:

Ideal for locations with limited manpower or remote areas.

Reduces dependency on physical presence.

8. Scalable:

Easily adaptable for use in schools, government buildings, or military bases.

Can be scaled for multiple flagpoles or centralized control.

6.3. Industrial Relevance

1. Demonstration of Automation Principles

Showcases scheduled and sensor-based automation, a key feature in industrial processes.

Similar to how machinery is automatically controlled in manufacturing and production lines.

2. IoT Integration

Uses Wi-Fi, cloud communication (e.g., MQTT/HTTP), and mobile control, which are widely used in Industrial IoT (IIoT) systems for remote equipment monitoring.

3. Remote Access and Control

Enables centralized control of operations, relevant for industries managing assets or machinery across multiple locations, such as:

Smart campuses

Industrial parks

Government facilities

4. Real-Time Monitoring and Alerts

Provides live status updates and error notifications, similar to condition monitoring in industrial automation and maintenance systems.

5. Scheduling and Time-Based Control

Incorporates RTC for time-triggered operations, commonly used in:

Power systems

Lighting controls

Industrial alarms

6. Scalability and Standardization

Can be scaled for multiple poles or automated flag systems across a region, much like industrial setups with multiple control points.

7. Educational and Prototyping Use

Serves as an educational prototype to understand embedded systems, motor control, and real-time IIoT applications-essential for industrial R&D and training.

6.4. Sustainable Development Goal

1. SDG 9 -Industry, Innovation, and Infrastructure

Promotes the use of innovative technologies and smart infrastructure.

Encourages IIoT and automation adoption in public and institutional spaces, showcasing technological progress.

Chapter 7

7.1. Conclusion

The IoT-based flag hoisting system demonstrates an effective integration of automation, wireless communication, and real-time control to modernize a traditional ceremonial process. By using components such as ESP32, motor drivers, limit switches, and real-time clocks, the system ensures safe, accurate, and scheduled hoisting of flags with minimal human intervention.

This project not only reduces manual effort but also enhances reliability, precision, and remote accessibility. It aligns with the principles of smart infrastructure, supports energy-efficient operation, and reflects real-world applications of IoT in civic and institutional automation. Overall, it stands as a valuable step toward smarter, more sustainable public services and provides a solid foundation for further IoT innovations.

7.2. Reference

1. Raj, R., & Singh, A. (2021). IoT-Based Automation System Using ESP8266 and Sensors. *International Journal of Engineering Research & Technology (IJERT)*, 10(06), 250–254.
2. Patil, S., & Khandekar, S. (2020). Smart Automation Using IoT: A Review. *International Research Journal of Engineering and Technology (IRJET)*, 7(4), 1429–1433.
3. Bharathi, A., & Kumar, R. (2019). Development of IoT Applications Using ESP32. *Journal of Emerging Technologies and Innovative Research (JETIR)*, 6(5), 101–105.
4. Kaur, P., & Sharma, R. (2018). Smart Flag Hoisting Using IoT. *Proceedings of the International Conference on Intelligent Computing and Control Systems (ICICCS)*, IEEE.
5. Ahmed, S., & Rao, P. (2019). Real-Time Embedded Systems for Automation Using IoT. *International Journal of Advanced Research in Computer Science*, 10(2), 567–572.
6. Arduino. (n.d.). ESP32 Documentation. Retrieved from: <https://docs.arduino.cc/>
7. MQTT.org. (n.d.). MQTT Protocol Overview. Retrieved from: <https://mqtt.org/>